SK6812 Bluetooth LED Controller - Complete System Documentation

System Overview

A professional Bluetooth-controlled LED lighting system that transforms standard SK6812 RGBW tape lights into intelligent, customizable lighting with smooth transitions and reliable wireless control.

Core Components

- ESP32 Microcontroller: Brain of the system, handles Bluetooth and LED control
- SK6812 RGBW LEDs: Smart pixels with individual white LED channels
- Android Control App: Simple interface for all lighting functions
- Professional Power System: Single 12V supply with buck converter for clean power

Core Lighting Features

Pure White Lighting

- True White LEDs: Uses only the dedicated white LED channel.
- No Color Mixing: White light comes from white diodes only (not RGB combination).
- Clean Output: Perfect for task lighting and accurate color rendering.

Smooth Transitions

- Fade Everything: All changes use smooth 2-second fades.
- No Jarring Changes: Gentle transitions between colors and effects.
- Professional Quality: Studio-grade lighting smoothness.

Color System

- 7 Solid Colors: Red, Orange, Yellow, Green, Blue, Indigo, Violet.
- Pure White: Dedicated white channel only.
- Color Accuracy: Precisely calibrated RGBW values.

Effects & Patterns

1. Color Shift Effect

- **Pattern:** All LEDs same color, slowly cycling through rainbow.
- **Timing:** 5 minutes per color, 10-second smooth transitions.
- **Flow:** Red → Orange → Yellow → Green → Blue → Indigo → Violet → Repeat.

2. Color Chase Effect

- **Pattern:** Flowing wave of color across the LED strip.
- **Smoothness:** No visible stepping, fluid motion.
- **Speed Control:** Adjustable wave speed via slider.

3. Color Palettes (Bonus Feature)

- **Sunset:** Orange - Red - Purple gradient.
- **Ocean:** Blue - Cyan - White water-inspired colors.
- **Forest:** Green - Yellow - Brown nature tones.
- **Rainbow:** Full spectrum color cycle.

Control Features

Bluetooth Connectivity

- **Range:** Up to 10 meters (30 feet).
- **Stability:** Auto-reconnect if connection drops.
- **Reliability:** Ring buffer system prevents lockups.
- **Compatibility:** Works with most Android devices.

Dimming & Speed Control

- **Brightness:** 0-100% smooth dimming.
- **Speed:** Adjust effect animation rates.
- **Real-time:** Instant response to control changes.

Power Management

- **Memory Function:** Remembers last state after power loss.
- **Auto Resume:** Restores previous settings when powered on.
- **Fade Protection:** Smooth power on/off to protect LEDs.

Advanced Features

Auto-Off Timer

- **Presets:** 15min, 30min, 1hr, 2hr options
- **Visual Feedback:** Countdown display in app
- **Smart Shutdown:** Smooth fade-out when timer completes

Professional Features

- **State Persistence:** Saves all settings automatically.
- **Error Handling:** Graceful recovery from communication issues.
- **Expandable:** Ready for 300+ LED expansion.
- **Debug Logging:** Detailed serial output for troubleshooting.

App Interface

Main Controls

- **Power Button:** Large, color-coded ON/OFF control.
- **Color Grid:** One-tap color selection.
- **Effect Buttons:** Instant pattern activation.
- **Sliders:** Smooth brightness and speed adjustment.

Visual Design

- **Dark Theme:** Easy on eyes in low light.
- **Color Coding:** Intuitive button colors match LED output.
- **Status Feedback:** Clear connection and timer status.

- **Material Design:** Professional Android interface.

Wiring setup

Components

- 12V power supply (main input).
- Buck converter (12V → 5V step-down).
- ESP32 microcontroller.
- DC12V SK6812 RGBW LED strip.
- Optional breakout board (for mounting and clean wiring).

Connection	From	To	Notes
+12V	12V power supply	LED strip +12V	Main LED power
+12V	12V power supply	Buck converter VIN	Step down to 5V
GND	12V power supply	LED strip GND	Common ground
GND	12V power supply	Buck converter GND	Common ground
5V	Buck converter VOUT	ESP32 5V/VIN	ESP32 power
GND	Buck converter GND	ESP32 GND	Common ground
GPIO18	ESP32	LED strip DIN	Data signal

ESP32 Flashing Setup

Step-by-Step Flashing Guide

1. Install Arduino IDE

- Download from **https://www.arduino.cc/en/software/#ide**

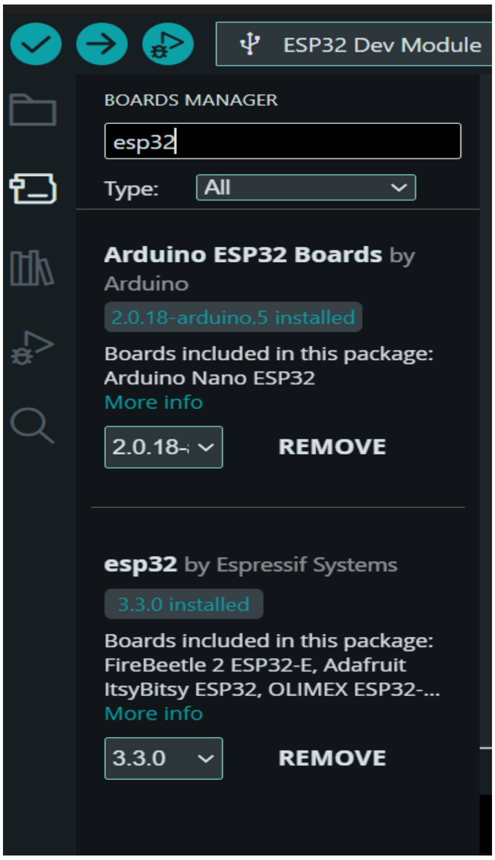
Install normally like any other program

2. Add ESP32 Board Support

1. Open Arduino IDE
2. Go to Tools → Board → Board Manager

3. Search for "ESP32"

4. Install "ESP32 by Espressif Systems" text



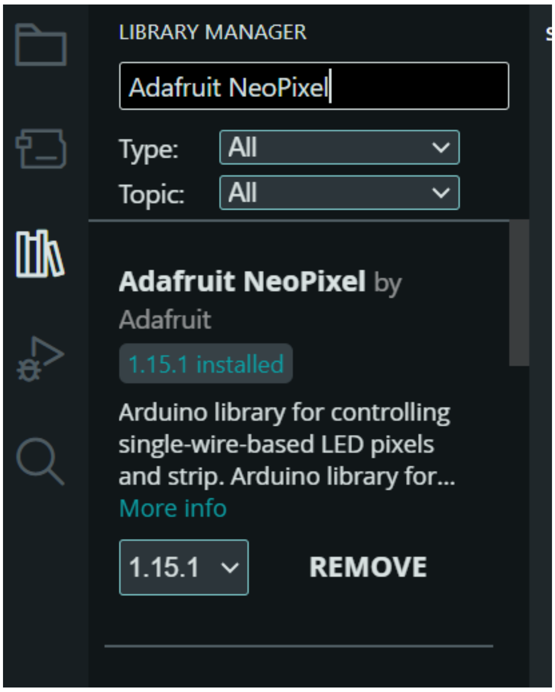
3. Install Required Libraries

1. Go to Tools → Manage Libraries

2. Search and install:

Adafruit NeoPixel by Adafruit

(Preferences library is built into ESP32)



5. Configure Board Settings

1. Tools → Board → ESP32 Dev Module
2. Tools → Port → Select your ESP32 COM port
3. Tools → Flash Frequency → 80MHz
4. Tools → Upload Speed → 115200
5. Tools → Partition Scheme → Default

6. Upload the Code

1. Open ESP32_Controller.ino file
2. Click Upload (right arrow button)
3. Wait for "Done uploading" message

App Installation & Bluetooth Setup

1. Install the App

- Copy LEDController.apk to your phone
- Open your file manager and tap the APK file

- Tap "Install" when prompted
- Open " LED Controller" app

2. Enable Bluetooth & Pair ESP32

- Go to **Phone Settings → Bluetooth → Turn ON**
- Look for **"ESP32_LED_Controller"** in available devices
- Tap **"ESP32_LED_Controller"** to pair
- Accept pairing request if prompted

3. Connect in App

- Open the LED Controller app
- Tap **"Connect to ESP32"**
- App will automatically find your paired ESP32
- Wait for "Connected: ESP32_LED_Controller" status
- You're ready to control lights!